Deterministic power law acceleration of optimistic gradient

Abstract

We prove deterministic last-iterate acceleration of ordinary optimistic gradient on finite-dimensional biaffine saddle problems. A binary radical-inverse power-law schedule gives a residual bound of order $T^{-1/\beta}$ for every fixed $3/2 < \beta \le 8/5$, with a sufficiently small explicit base step. In particular, the exponent $33/50$ is valid with a concrete fixed base. The schedule, scalar logarithmic integral and derandomization principle are based on prior extragradient work. The new transfer controls the ordered optimistic-gradient product by pairing each large step with a neighboring base step in a frequency-dependent energy. Its additional logarithmic error is a monotone tail moment and can be absorbed uniformly. We include a self-contained special-case quadrature proof and treat every prefix, zero singular values and the actual last iterate. The result uses the unchanged optimistic-gradient recurrence and makes no assertion about the larger base step used in the motivating experiment or about optimality among optimistic-gradient schedules.

Keywords: optimistic gradient; last iterate; biaffine saddle problems; power law stepsizes; deterministic acceleration

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1 The recurrence and the accelerated bound

Consider a biaffine function

$$\ell(x,y) = x^{\mathsf{T}}Ay + u^{\mathsf{T}}x + v^{\mathsf{T}}y + c, \qquad G(x,y) = (Ay+u, -A^{\mathsf{T}}x - v), \tag{1}$$

where $x \in \mathbb{R}^{d_1}$, $y \in \mathbb{R}^{d_2}$, $\|A\| \le L$ and $L > 0$. Assume that a saddle point $z_* = (x_*, y_*)$ exists, equivalently $G(z_*) = 0$. We study

$$w_{-1} = z_0, \qquad w_t = z_t - \eta_t G(w_{t-1}), \qquad z_{t+1} = z_t - \eta_t G(w_t), \qquad t \ge 0. \tag{2}$$

The output is the actual last iterate z_T and the criterion is $\|G(z_T)\|$. The same infinite deterministic schedule is used at every horizon.

For an integer $t = \sum_{j\ge 0} \varepsilon_j 2^j$, define its binary radical inverse by

$$\phi_t = \sum_{j\ge 0} \varepsilon_j 2^{-j-1}, \qquad \phi_0 = 0. \tag{3}$$

Fix

$$\frac{3}{2} < \beta \le \frac{8}{5}, \qquad p = \frac{4}{29}, \qquad q = 2p = \frac{8}{29}, \qquad \gamma_\beta = -\frac{2\pi\cos(\pi\beta/3)}{\sin(\pi\beta/2)} > 0, \tag{4}$$

and put

$$h_\beta = \min\left\{\frac{1}{2}, \left(\frac{\gamma_\beta(2-\beta)}{6\beta}\right)^{1/(2-\beta)}, \left(\frac{q\gamma_\beta}{204}\right)^{1/(5-\beta)}, \frac{\gamma_\beta(4-\beta)}{816}\right\}. \tag{5}$$

All four terms are positive. For a fixed $0 < h \le h_\beta$, use

$$\eta_t = \frac{h}{L}\max\left\{1, \left(\frac{p}{1-\phi_t}\right)^{1/\beta}\right\}. \tag{6}$$

Theorem 1. *For every problem and initialization in* (1)–(2)*, the schedule* (6) *satisfies*

$$\|G(z_T)\| \le C_{\beta,h} L \|z_0 - z_*\| \, T^{-1/\beta}, \qquad T \ge 1, \tag{7}$$

where one may take

$$C_{\beta,h} = \max\left\{1, \frac{\mathrm{e}^{29}}{h}\left(\frac{12}{\mathrm{e}\beta q\gamma_\beta}\right)^{1/\beta}\right\}. \tag{8}$$

The constant is independent of the dimensions, matrix, affine terms, initial point, saddle point and horizon. The exact values $\beta = 50/33$ *and* $h = 2^{-20}$ *are admissible, giving exponent* $33/50$.

The constants are conservative. For every $0 < \varepsilon \le 1/24$, choosing $\beta = (2/3 - \varepsilon)^{-1}$ gives order $T^{-2/3+\varepsilon}$. The endpoint exponent $2/3$ is not included. Neither the base $h = 1/2$ in the motivating experiment nor an optimal rate for optimistic gradient is asserted.

Wu, Zheng, Cai and Luo [1] introduce power-law stepsizes and binary radical-inverse derandomization for extragradient. Their public version proves deterministic extragradient acceleration and, in Appendix E, a weaker randomized geometric-mean result for optimistic gradient. That appendix explicitly leaves the sharper optimistic-gradient analysis open. The schedule family, the scalar Mellin integral and the scalar quadrature estimate used here belong to that prior work. We give an elementary proof of the needed quadrature special case for completeness. The contribution is the deterministic transfer to the noncommuting product generated by (2), at the explicit smaller base (5).

All comparisons and source numbering refer to arXiv:2606.01764v1. A distinct accepted text was not available for comparison. Constant-step final-residual estimates for optimistic gradient on general monotone Lipschitz problems are established in [2]; the present theorem is restricted to biaffine problems and changes only the step schedule. In particular, the pairing below is an estimate on the original ordered trajectory, not a reset of the method.

2 Scalar estimates for the schedule

Write

$$E(r) = 1 - r^2 + r^4. \tag{9}$$

For $0 < b \le 1/2$, let R have the mixture distribution

$$(1-q)\delta_b + q\,\mathrm{Pareto}(b,\beta). \tag{10}$$

Thus its quantile is $r(u) = b\max\{1, [q/(1-u)]^{1/\beta}\}$, $0 \le u < 1$.

Lemma 1. *For the parameters above,*

$$\mathbb{E}\log E(R) \le -q\gamma_\beta b^\beta + \frac{q\beta}{2-\beta}b^2, \tag{11}$$

$$\mathbb{E}\min(R^4, 1) \le b^4 + \frac{4q}{4-\beta}b^\beta. \tag{12}$$

Proof. The scalar integral computed in [1, Lemma A.3 and Appendix C.1] is

$$J_\beta = \int_0^\infty \log E(r)\, r^{-\beta-1}\, dr = \frac{2\pi\cos(\pi\beta/3)}{\beta\sin(\pi\beta/2)} = -\frac{\gamma_\beta}{\beta}. \tag{13}$$

It is absolutely convergent for $0 < \beta < 2$. For example, integration by parts and $s = r^2$ give $\beta J_\beta = \int_0^\infty s^{-\beta/2}(2s-1)/(s^2-s+1)\,ds$; factoring the quadratic and applying the Euler integral gives (13).

For $t \ge 0$, the derivative of $\log(1-t+t^2)+t$ is $t(1+t)/(1-t+t^2) \ge 0$. Hence $\log E(r) \ge -r^2$. Also $\log E(b) \le 0$ for $b \le 1$. Integrating the Pareto density yields

$$\begin{aligned}\mathbb{E}\log E(R) &\le q\beta b^\beta \int_b^\infty \log E(r) r^{-\beta-1}\, dr \\ &\le q\beta b^\beta \left(J_\beta + \frac{b^{2-\beta}}{2-\beta}\right),\end{aligned}$$

which proves (11). Direct integration at the cutoff $r = 1$ gives

$$\mathbb{E}\min(R^4,1) = (1-q)b^4 + q\left(\frac{4}{4-\beta}b^\beta - \frac{\beta}{4-\beta}b^4\right), \tag{14}$$

and (12) follows. ∎

Lemma 2. *For every integer $n \ge 1$, with $r_j = r(\phi_j)$,*

$$\sum_{j=0}^{n-1} \log E(r_j) \le n\mathbb{E}\log E(R) + 58, \tag{15}$$

$$\sum_{j=0}^{n-1} \min(r_j^4, 1) \le n\mathbb{E}\min(R^4,1). \tag{16}$$

The first estimate is a special case of [1, Theorem D.16]; the second uses its monotonicity principle, Theorem D.5. The following proof supplies a uniform constant in the particular parameter range needed here.

Proof. We first record the left-bias property of (3). If $F_n(x) = n^{-1}\#\{j < n : \phi_j < x\}$, then $F_n(x) \geq x$ on $[0,1]$. This follows by induction from $\phi_{2j} = \phi_j/2$ and $\phi_{2j+1} = (1+\phi_j)/2$. For $n = 2m$, apply the induction hypothesis on each half interval. For $n = 2m+1$, the lower half has $m+1$ points and the upper half has m; the resulting lower bounds are $2(m+1)x/(2m+1)$ below $1/2$ and $(1+2mx)/(2m+1)$ above it. Both are at least x. Consequently every integrable nondecreasing H satisfies

$$\sum_{j=0}^{n-1} H(\phi_j) \leq n \int_0^1 H(u)\,du. \tag{17}$$

Truncation covers integrable unbounded functions. This already proves (16).

Put $\alpha = 2/\beta \in [5/4, 4/3)$ and $\lambda(s) = \log(1 - s + s^2)$ for $0 \leq s \leq 1$. Define

$$\Psi(t) = \begin{cases} \lambda(t^\alpha), & 0 \leq t \leq 1, \\ \lambda(t^{-\alpha}), & t \geq 1. \end{cases} \tag{18}$$

The bounds $\Psi \leq 0$, $|\lambda(s)| \leq s$, $|\lambda'(s)| \leq 4/3$ and $|\lambda''(s)| \leq 3$ give

$$\int_0^\infty |\Psi(t)|\,dt \leq 5, \qquad \mathrm{Var}(\Psi') \leq \frac{8\alpha}{3} + \frac{3\alpha^2}{2\alpha - 1} + \frac{3\alpha^2}{2\alpha + 1} + 2\alpha < 12. \tag{19}$$

Here the variation includes the jump of size 2α in Ψ' at 1. The displayed bounds follow by differentiating on the two sides and integrating the resulting powers. For example, bounding the three contributions by $16/3$, $208/63$ and $8/3$ gives $712/63 < 12$.

For $0 < \sigma \leq 1$, let $f_\sigma(u) = \Psi((1-u)/\sigma)$. It is nonpositive, $f_\sigma(1) = 0$, $f_\sigma(0) \leq 0$, and

$$\int_0^1 |f_\sigma| \leq 5\sigma, \qquad \mathrm{Var}(f_\sigma') \leq 12/\sigma. \tag{20}$$

For a grid of size m and a shift $0 \leq \theta \leq 1/2$, define $S = \sum_{j<m} f_\sigma((j+\theta)/m) - m\int_0^1 f_\sigma$. The sign of f_σ gives $S \leq 5m\sigma$. For an absolutely continuous f with derivative of bounded variation, comparison with the linear interpolant on each grid cell gives

$$\left| \sum_{j<m} f((j+\theta)/m) - m\int_0^1 f - (\theta - \tfrac{1}{2})(f(1) - f(0)) \right| \leq \frac{2}{m}\,\mathrm{Var}(f'). \tag{21}$$

Indeed, on a cell of width $1/m$, the interpolant's slope is the average of f'. Both the shifted-value error and the normalized integral error are at most m^{-1} times that cell's derivative variation. Summing proves (21). Its boundary term is nonpositive for f_σ, so

$$S \leq \min\{5m\sigma, 24/(m\sigma)\}. \tag{22}$$

Split the first n integers into aligned consecutive blocks with lengths given by the decreasing powers of two in the binary expansion of n. In a block of length $m = 2^k$ beginning at s, the lower $k+1$ bits of s vanish. The multiset of radical inverses in that

block is $\{(j+\theta)/m : 0 \le j < m\}$ with $\theta = m\phi_s < 1/2$. There is at most one block of each dyadic size. Summing (22), using its first bound when $m\sigma \le 1$ and its second otherwise, gives

$$\sum_{j<n} f_\sigma(\phi_j) - n\int_0^1 f_\sigma \le 10 + 48 = 58. \tag{23}$$

Finally set $\widehat{r}(u) = b[q/(1-u)]^{1/\beta}$ and $\sigma = qb^\beta$. On $u \le 1-q$, the difference $\log E(r(u)) - \log E(\widehat{r}(u))$ is nondecreasing, since $\widehat{r} \le b \le 1/2$ and E is decreasing there; after that interval the difference is zero. Also $\max\{0, 4\log\widehat{r}(u)\}$ is nondecreasing. The identity $E(r) = r^4 E(1/r)$ shows that

$$\log E(\widehat{r}(u)) - \max\{0, 4\log\widehat{r}(u)\} = f_\sigma(u). \tag{24}$$

Both removed monotone terms have nonpositive quadrature error by (17). Equation (23) therefore proves (15). ∎

3 A paired energy for the ordered optimistic product

For a singular value $a \in (0, L]$, identify its two-dimensional real block with a complex coordinate. Up to complex conjugation, the operator is multiplication by $\mathrm{i}a$. If $d_t = w_{t-1} - z_t$, one update of (2) with $r = a\eta_t$ becomes

$$\begin{pmatrix} d_{t+1} \\ z_{t+1} \end{pmatrix} = M(r)\begin{pmatrix} d_t \\ z_t \end{pmatrix}, \qquad M(r) = \begin{pmatrix} r^2 - \mathrm{i}r & r^2 \\ -r^2 & 1 - \mathrm{i}r - r^2 \end{pmatrix}. \tag{25}$$

Here $d_0 = 0$. The matrix representation is the one in [1, Lemma E.1]. The following estimate uses a fixed metric for each frequency throughout its entire run.

Lemma 3. *Let $0 < b \le 1/2$ and $r \ge b$. With $P_b = \mathrm{diag}(b^{-1/2}, 1)$,*

$$\|P_b M(b) M(r) P_b^{-1}\| \le \sqrt{E(r)}\exp\big(17b\min(r^4, 1)\big). \tag{26}$$

The initial base step satisfies

$$\|P_b M(b)(0,1)^{\mathsf{T}}\| \le 1. \tag{27}$$

The Euclidean norm of the second row of $M(r)P_b^{-1}$ satisfies the right side of (26) as well.

Proof. Writing $B = P_b M(b) M(r) P_b^{-1}$, direct multiplication gives

$$\begin{aligned} B_{11} &= -br\big(1 + \mathrm{i}(b+r)\big), & B_{12} &= b^{3/2} - \mathrm{i}\sqrt{b}\, r(b+r), \\ B_{21} &= \sqrt{b}\big(-r^2 + \mathrm{i}br(b+r)\big), & B_{22} &= 1 - \mathrm{i}(b+r) - (b^2 + r^2 + br) + \mathrm{i}(br^2 + b^2 r). \end{aligned}$$

The squared Frobenius norm obeys the exact identity

$$\begin{aligned} \|B\|_{\mathrm{F}}^2 - E(r) = {} & b^5 r^2 + 2b^4 r^3 + 2b^4 r^2 + b^4 + b^3 r^4 + 4b^3 r^3 + b^3 r^2 + b^3 \\ & + 2b^2 r^4 + 2b^2 r^3 - b^2 + 2br^4. \end{aligned} \tag{28}$$

The terms independent of r sum to $b^2(b^2+b-1) \le 0$. Since $r \ge b$ and $b \le 1$, each of the other nine monomials is bounded by its numerical coefficient times br^4. Their coefficients sum to 17. Therefore

$$\|B\|^2 \le \|B\|_{\mathrm{F}}^2 \le E(r) + 17br^4. \tag{29}$$

The inequality $E(r) \ge (1+r^4)/2$ gives $r^4/E(r) \le 2\min(r^4, 1)$. Applying $\sqrt{1+x} \le \exp(x/2)$ proves (26).

For the initial step, direct calculation yields

$$\|P_b M(b)(0,1)^{\mathsf{T}}\|^2 = 1 - b^2 + b^3 + b^4 \le 1. \tag{30}$$

The squared norm of the final output row is $E(r) + br^4$. The same argument proves its bound, even with 1 in place of 17 in the exponent. ∎

The metric depends on the frequency through $b = ha/L$, and its first weight grows as $a \downarrow 0$. Equations (27) and (29) are precisely what avoid any resulting nonuniform initialization or output constant. No positive lower bound on a nonzero singular value is used.

4 Every prefix and the uniform rate

Proof of Theorem 1. Translate the variables by z_*. The affine operator becomes $G(z) = J(z - z_*)$, where $J = \left(\begin{smallmatrix} 0 & A \\ -A^{\mathsf{T}} & 0 \end{smallmatrix}\right)$. An orthogonal singular-value decomposition of A separates the recurrence into complex scalar blocks of frequencies $a \in (0, L]$ and kernel coordinates. The latter have zero residual. It suffices to bound every nonzero block uniformly.

Fix such a block and set $b = ha/L \le h_\beta \le 1/2$. The identities for the radical inverse imply

$$a\eta_{2j} = b, \qquad a\eta_{2j+1} = r_j = b \max\left\{1, \left(\frac{q}{1-\phi_j}\right)^{1/\beta}\right\}. \tag{31}$$

Indeed, every even quantile is below $1/2 < 1 - p$, and on an odd index the tail mass doubles from p to $q = 2p$.

The first step is $M(b)$ and obeys (27). A large step $M(r_j)$ is then followed by a base step $M(b)$. For $T = 2n+1$, all n such pairs are complete. For $T = 2n$, the first $n-1$ pairs are complete and the final large step is handled by the output-row bound in Lemma 3. The second component of a weighted state has norm at most that of the state. Thus in both cases, writing ζ_t for this scalar block,

$$|\zeta_T| \le |\zeta_0| \prod_{j=0}^{n-1} \sqrt{E(r_j)} \exp\left(17b \sum_{j=0}^{n-1} \min(r_j^4, 1)\right), \qquad n = \lfloor T/2 \rfloor, \tag{32}$$

for every $T \ge 2$. This preserves the actual product order and includes both terminal parities.

Lemmas 1 and 2 bound the logarithm of the factor multiplying $|\zeta_0|$ by

$$29 + nb^\beta \left[-\frac{q\gamma_\beta}{2} + \frac{q\beta}{2(2-\beta)} b^{2-\beta} + 17b^{5-\beta} + \frac{68q}{4-\beta} b \right]. \tag{33}$$

The last three terms in brackets are each at most $q\gamma_\beta/12$, by (5) and $b \le h \le h_\beta$. Hence

$$|\zeta_T| \le \mathrm{e}^{29} \exp\left(-\frac{q\gamma_\beta}{4} n(ha/L)^\beta \right) |\zeta_0|. \tag{34}$$

For $T \ge 2$, $n \ge T/3$. Maximizing $x \exp(-cx^\beta)$ over $x > 0$ gives $(\mathrm{e}\beta c)^{-1/\beta}$. Applying this with $x = a/L$ proves

$$a|\zeta_T| \le \frac{\mathrm{e}^{29}}{h} \left(\frac{12}{\mathrm{e}\beta q \gamma_\beta} \right)^{1/\beta} LT^{-1/\beta} |\zeta_0|. \tag{35}$$

For $T = 1$, equation (27) gives $a|\zeta_1| \le L|\zeta_0|$. Squaring the same uniform estimate for every nonzero singular block and summing proves (7). Kernel components contribute nothing to the residual and do not require the metric P_b.

Finally, when $\beta = 50/33$,

$$\gamma_\beta = \frac{2\pi \sin(\pi/198)}{\sin(8\pi/33)} \ge \frac{2\pi}{99} > \frac{2}{33}. \tag{36}$$

We used $\sin x \ge 2x/\pi$ on $[0, \pi/2]$ and $\pi > 3$. For $h = 2^{-20}$,

$$\begin{aligned} &h^{2-\beta} = 2^{-320/33} < \frac{1}{512} < \frac{8}{2475} = \frac{(2/33)(2-\beta)}{6\beta}, \\ &h^{5-\beta} \le h < \frac{q(2/33)}{204}, \qquad h < \frac{(2/33)(4-\beta)}{816}, \qquad h \le \frac{1}{2}. \end{aligned} \tag{37}$$

These are the four constraints in (5), proving the exact specialization. ∎

The theorem establishes an upper bound for a fixed infinite schedule and the actual last z-iterate. It does not transfer the extragradient lower bound in [1] to optimistic gradient. The smaller base and the constant in (8) establish an asymptotic rate; they are not presented as practical tuning recommendations. The argument uses the spectral structure of a biaffine problem and does not establish the same claim for general nonlinear monotone operators.

References

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